DOI: https://doi.org/10.31640/LS-2025-3-01

UDC 616.61-008.64-07:577.1

L. D. Denova, *MD, PhD-student,*
ORCID ID: 0000-0002-5678-5885,
marbua18@gmail.com

S. V. Kushnirenko, *MD, Doctor of Medical Sciences, Professor,*
ORCID ID: 0000-0001-5518-7210,
stella-alex@i.ua

Shupyk National Healthcare University of Ukraine, Kyiv, Ukraine,
Department of Nephrology and Renal Replacement Therapy

EVALUATION OF THE UROMODULIN PROFILE AND CLASSICAL MARKERS IN RISK STRATIFICATION FOR PROGRESSION OF CHRONIC KIDNEY DISEASE STAGES 1–3

Abstract. *This study presents a comparative analysis of the analytical informativeness of indices within the uromodulin profile–fractional excretion of uromodulin (FeUmod), fractional secretion of uromodulin (FsUmod), and serum uromodulin (sUmod) – as well as classical markers of kidney function: serum creatinine (sCrea), estimated glomerular filtration rate (eGFR) by the CKD-EPI formula, albuminuria (uAlb), and the albumin-to-creatinine ratio (uAlb/uCrea) in patients with chronic kidney disease (CKD) stages 1–3.* ***Aim of the study.*** *To assess the diagnostic performance of uromodulin profile indices (FeUmod, FsUmod, sUmod) compared to classical markers of kidney function (sCrea, eGFR, uAlb, uAlb/uCrea) in patients with CKD stages 1–3.* ***Materials and methods.*** *The study included 91 patients with CKD stages 1–3. Biomarker levels were analyzed at three time points: T0 (baseline), T1 (after 3 months), and T2 (after 6 months). Correlation analysis was performed to assess associations between biomarkers. The discriminatory ability to detect reduced eGFR < 60 mL/min/1.73 m² was evaluated using ROC analysis. Cluster analysis was employed to identify phenotypes of renal functional status.* ***Results.*** *At the final time point (T2), FeUmod demonstrated the highest discriminatory performance, surpassing sCrea, FsUmod, sUmod, and uAlb/uCrea. In correlation analysis, FeUmod also showed the strongest association with eGFR. Cluster analysis identified three distinct phenotypes of patients characterized by different patterns of tubular and glomerular dysfunction.* ***Conclusions.*** *Indices of the uromodulin profile, particularly FeUmod, exceed classical markers of kidney function in their ability to stratify the risk of decreased renal filtration capacity in patients with CKD stages 1–3. Incorporating FeUmod into laboratory monitoring tools in nephrology and general clinical practice is advisable.*

Keywords: *uromodulin; fractional excretion of uromodulin; fractional secretion of uromodulin; chronic kidney disease; estimated glomerular filtration rate; risk stratification; biomarkers.*

INTRODUCTION

CKD remains one of the leading global challenges in modern nephrology. CKD progression often occurs latently, and traditional markers–particularly sCrea and eGFR–do not always reflect early

© Л. Д. Денова, С. В. Кушніренко, 2025

changes in nephron function, especially during the pre-dialysis period [1, 2, 3]. Therefore, increasing attention is being paid to tubular-origin biomarkers, among which Umod is considered a promising tool for the early detection of nephron dysfunction [4, 7, 8].

Umod (Tamm–Horsfall protein), the most abundant urinary protein in healthy individuals, is synthesized in the distal segments of the nephron and plays a protective role in the urinary tract. It is also involved in electrolyte homeostasis and the metabolic adaptation of the tubular epithelium [8, 9 10, 11]. Impairment in Umod biosynthesis or excretion is associated with both CKD progression and the development of acute kidney injury (AKI) [5, 12, 13]. The serum level of sUmod, as well as its derivative indices–FeUmod and FsUmod – are increasingly regarded as integrative markers of nephron functional reserve and distal tubular damage [6, 7, 8, 13, 14].

Recent molecular and genetic studies have shown that Umod splicing variants and genetic polymorphisms influence renal epithelial adaptation to stress and may also modify the clinical course of kidney diseases [9, 15]. In particular, multi-omics approaches allow for the identification of novel therapeutic targets in CKD treatment, including those involving Umod and related proteins [12, 16]. It has also been demonstrated that Umod levels may vary depending on age, metabolic status, and cardiovascular risk [17, 18, 19]. Meanwhile, modern analytical platforms, including artificial intelligence models, enhance the ability to predict CKD progression based on integrated biomarker data [20].

Given the above, comparing the diagnostic value of Umod profile indices with classical kidney function markers for the purpose of CKD risk stratification is a timely and relevant research objective.

AIM

To evaluate the diagnostic significance of Umod profile indices (FeUmod, FsUmod, sUmod) in comparison with classical markers of kidney function (sCrea, eGFR, uAlb, and uAlb/uCrea) in patients with CKD stages 1–3 for the purpose of risk stratification of decreased renal filtration capacity.

MATERIALS AND METHODS

This prospective, randomized study was conducted between 2021 and 2023 at VETA-PLUS LLC (Brovary, Ukraine) and included 91 patients with CKD stages 1–3. The diagnosis was established according to KDIGO criteria (2012, 2021) using the CKD-EPI formula. Patients were randomized into four groups: two main groups received standard therapy in combination with glutathione (GSH) (100 mg twice daily) or coenzyme Q10 (CoQ10) (100 mg daily); the third group received standard therapy only; the fourth group served as the control. GSH and CoQ10 were administered in the form of dietary supplements.

Biomarker parameters (FeUmod, FsUmod, sUmod, sCrea, eGFR, uAlb, and uAlb/uCrea) were assessed at three time points: before treatment (T0), after 3 months (T1), and after 6 months (T2). Correlation analysis, ROC analysis, and clustering were used to evaluate associations and diagnostic performance. The study was approved by the Ethics Committee of the Shupyk National Healthcare University of Ukraine (Protocol No. 12, dated November 29, 2021), and all participants provided written informed consent.

RESULTS

At the T2 time point, key kidney function biomarkers were evaluated in patients with CKD stages 1–3. The mean FeUmod level was 6.63 ± 7.13, with a wide range of variation (0.35–63.28), indicating heterogeneity in the tubular response among participants. The FsUmod level was extremely low

(<0.0001 in most cases). The mean serum uromodulin (sUmod) concentration was 57.6 ± 10.2 ng/mL. The mean sCrea level was 81.4 ± 14.6 μmol/L, and the eGFR was 85.4 ± 16.1 mL/min/1.73 m². The data are presented in Table 1.

Table 1. Descriptive statistics of kidney function biomarkers in patients with CKD stages 1–3 (T2 time point)

№	Variable	n	Mean	SD	Min	Q1	Median	Q3	Max
1	FeUmod	91	6.63	7.13	0.35	3.14	4.85	8.6	63.28
2	FsUmod	91	0*	0.01	0*	0*	0*	0*	0,06
3	sUmod	91	57.59	10.15	40.6	51	56.4	61.75	93.4
4	sCrea	91	81.4	14.61	54	74	78	86	175
5	eGFR	91	85.38	16.11	37.5	75.35	82.54	97.36	126.11
6	uAlb	91	23.28	13.74	4.3	16.3	22.6	26.8	92.9
7	uAlb/uCrea	91	3.18	2.65	0.3	1.46	2.56	4.13	17.53

Note:

data are presented as mean ± standard deviation, as well as median, quartiles, minimum, and maximum;

** the FsUmod values exhibited extremely low absolute levels (< 0.0001);*

n – number of patients.

These baseline characteristics indicate the presence of both compensated forms of CKD (elevated eGFR values) and patients with signs of disease progression (elevated sCrea, increased uAlb, wide dispersion in FeUmod).

Based on FeUmod, FsUmod, sUmod, sCrea, eGFR, uAlb, and uAlb/uCrea, k-means cluster analysis identified three phenotypically distinct clusters of patients with CKD stages 1–3. The results are presented in Table 2.

Table 2. Mean values of kidney function biomarkers in patients from different phenotypic clusters (T2)

№	Cluster	FeUmod	FsUmod	sUmod	sCrea	eGFR	uAlb	uAlb/uCrea	n
1	0	0.74	0.06	67.6	101.5	65.28	25.55	5.34	2
2	1	5.97	0*	57.61	79.54	87.01	21.21	2.71	85
3	2	23.78	0.01	52.12	110.75	60.86	65.97	12.03	4

Note:

clustering was performed using the k-means method based on the biomarkers FeUmod, FsUmod, sUmod, sCrea, eGFR, uAlb, and uAlb/uCrea;

** FsUmod values in Cluster 1 had extremely low absolute values (<0.0001);*

n – number of patients in each cluster.

The phenotype of Cluster 0 (n = 2) was characterized by elevated sCrea, decreased eGFR, and increased uAlb, which may indicate progressive glomerular-tubular dysfunction. Cluster 1 (n = 85)

represented the majority of patients with moderate biomarker values. Cluster 2 (n = 4) exhibited extremely high FeUmod and uAlb/uCrea levels, suggesting active nephropathy with a predominant tubular component.

Thus, cluster analysis enabled the identification of functional phenotypes among patients with CKD stages 1–3, opening prospects for a personalized approach to risk stratification.

To visualize the phenotypic differentiation of patients, a Principal Component Analysis (PCA) was conducted based on kidney function biomarkers. The constructed PCA model displayed the distribution of patients along the first two principal components (PC1 and PC2), which together explain more than 70% of the total variance. Figure 1 clearly illustrates the separation of three clusters, confirming the validity of the clustering and reflecting differences in the structure of the Umod profile.

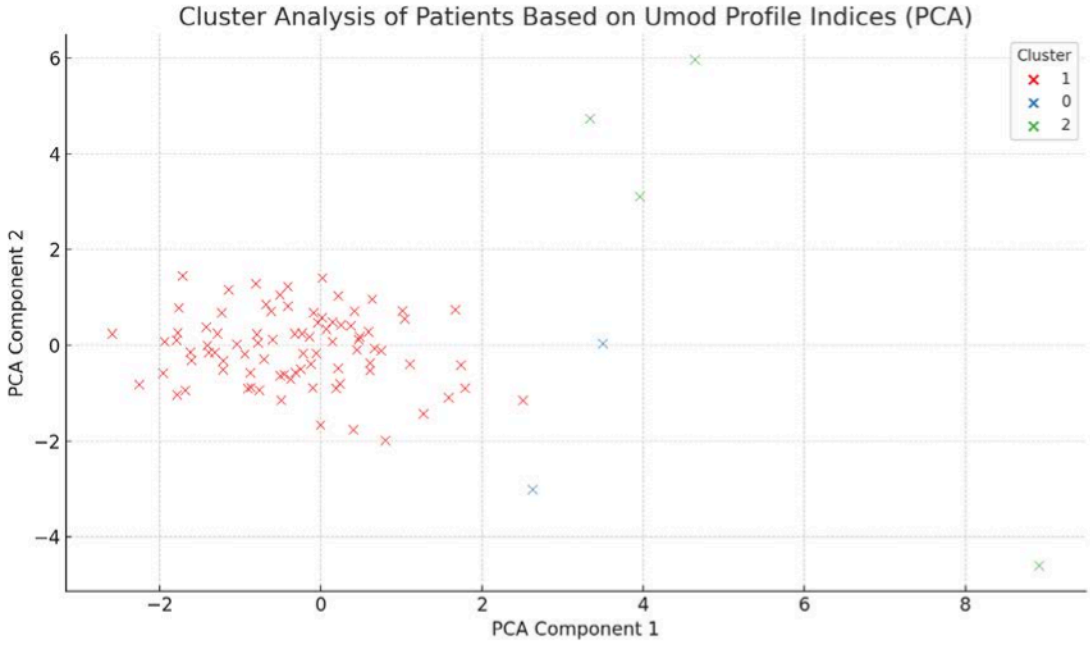

Figure 1. Distribution of patients with CKD stages 1–3 based on PCA cluster analysis
Note: each point represents an individual patient; clusters are color-coded according to group membership (Cluster 0 – blue, Cluster 1 – red, Cluster 2 – green).

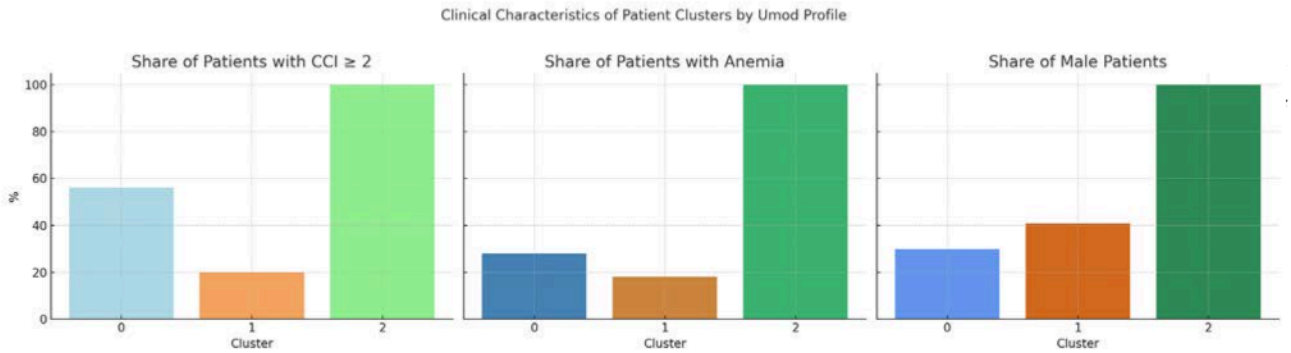

Figure 2. Frequency of Charlson Comorbidity Index (CCI) ≥ 2, anemia, and male sex among patients in each of the three clusters identified through biomarker-based clustering

To assess the diagnostic value of kidney function biomarkers, a ROC analysis was performed to detect eGFR < 60 mL/min/1.73 m² (see Figure 3).

The highest AUC values were observed for FeUmod (AUC = 0.985), sCrea (AUC = 0.977), and uAlb/uCrea (AUC = 0.955), demonstrating their high and comparable informativeness in detecting reduced eGFR.

The slight advantage of FeUmod may reflect its sensitivity to early tubular impairment; however, classical kidney function markers remain reliable diagnostic tools.

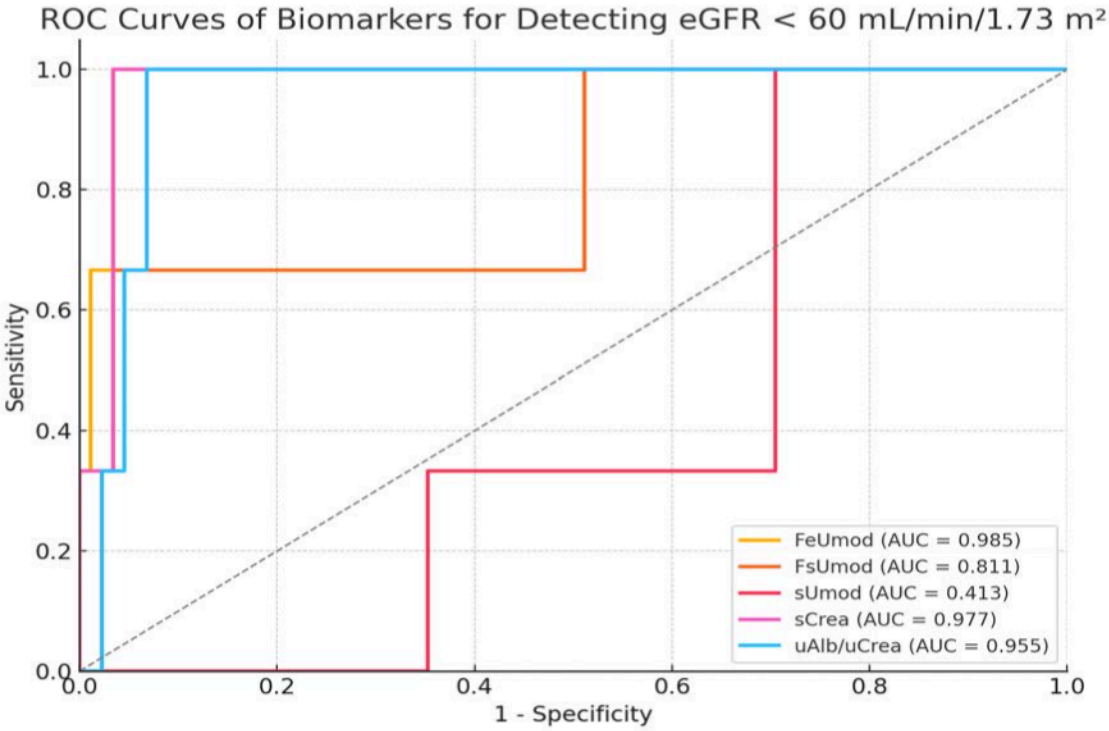

Figure 3. ROC curves of kidney function biomarkers (FeUmod, FsUmod, sUmod, sCrea, uAlb/uCrea) for detecting a decrease in eGFR < 60 mL/min/1.73 m²

Thus, indices of the uromodulin profile, particularly FeUmod, demonstrate high diagnostic informativeness and may be useful in identifying patients at risk of declining renal filtration capacity.

To enhance sensitivity to changes in both tubular and glomerular function, a composite index FeUmod × sCrea was constructed. ROC analysis of this index revealed exceptionally high discriminatory ability for detecting eGFR < 60 mL/min/1.73 m² – AUC = 0.996, see Figure 4.

The construction of the index was based on the assumption that combining markers of tubular origin (FeUmod) and glomerular origin (sCrea) would enhance the discriminatory capacity for detecting reduced filtration function. Thus, the combination of markers of tubular and glomerular origin provides near-optimal informativeness and can be used as an integrative tool for risk stratification in patients with CKD stages 1–3.

To confirm the independent association between FeUmod and renal filtration function, a multiple linear regression model was constructed with eGFR as the dependent variable. The model included FeUmod (T2), sCrea (T2), age, and sex (see Figure 5).

FeUmod remained an independent predictor of eGFR after adjustment for age, sex, and sCrea ($\beta = -0.44$; $p = 0.038$), indicating its autonomous prognostic value. The model demonstrated an explanatory power of $R^2 = 0.577$.

Thus, FeUmod demonstrated independent prognostic value for eGFR, even within a model adjusted for the main clinical characteristics of the patients.

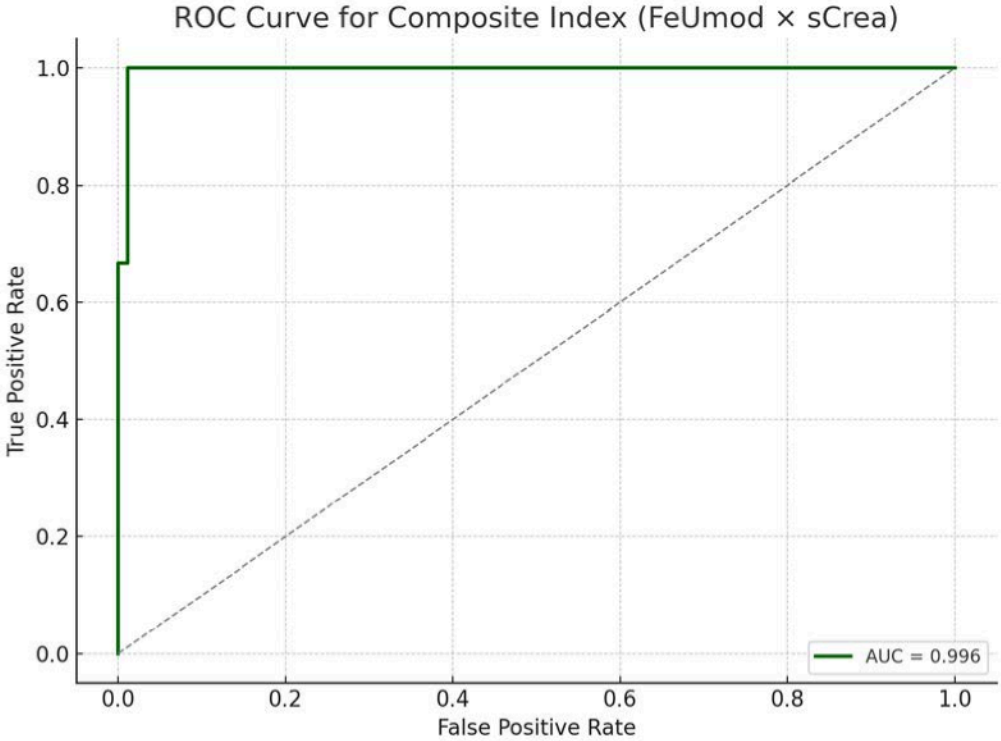

Figure 4. ROC curve for the composite index FeUmod × sCrea in detecting a reduction in eGFR < 60 mL/min/1.73 m²

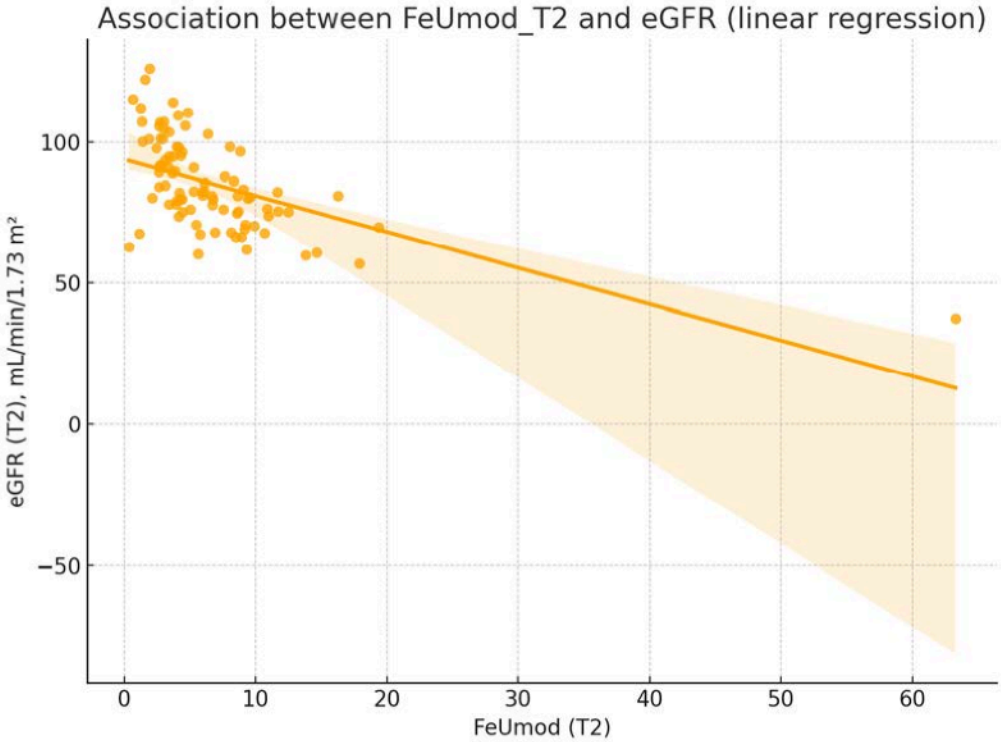

Figure 5. Association between FeUmod (T2) level and eGFR (T2) in patients with CKD stages 1–3

Note: Constructed using a linear regression model (β = –0.44; p = 0.038).

DISCUSSION

In the present study, the diagnostic informativeness of Umod profile indices was evaluated in patients with CKD stages 1–3 compared to classical markers of kidney function. The obtained results indicate a high discriminatory ability of FeUmod, which–on par with traditional markers (sCrea, uAlb/uCrea)–was significantly associated with decreased eGFR. Furthermore, cluster analysis revealed heterogeneity in patient phenotypes based on uromodulin-related alterations, opening new perspectives for risk stratification.

The results of this study demonstrated the clinical feasibility of incorporating uromodulin profile indices, particularly FeUmod, into the laboratory monitoring of patients with CKD stages 1–3. It was shown that the diagnostic performance of FeUmod in detecting eGFR < 60 mL/min/1.73 m² was comparable to that of classical markers such as sCrea and uAlb/uCrea. These findings are consistent with previous studies that highlighted Umod's sensitivity to early tubular alterations [5, 6, 8].

In this study, for the first time in clinical practice, a parallel ROC curve analysis was conducted for FeUmod, sCrea, uAlb/uCrea, sUmod, and FsUmod within a single cohort of CKD patients, enabling direct comparison of their diagnostic performance. The AUC values for FeUmod (0.985), sCrea (0.977), and uAlb/uCrea (0.955) confirmed their high sensitivity in detecting reduced filtration capacity, echoing findings from studies where both serum and urinary Umod were associated with nephron functional reserve [6, 8].

Additional analytical value was provided by the cluster analysis, which revealed three phenotypic subgroups of patients differing in the extent of tubular and glomerular damage. It was observed that the most clinically burdened patients (cluster 2) exhibited the highest FeUmod and FsUmod levels, as well as the highest prevalence of anemia and elevated CCI, confirming the association between tubular dysfunction and overall somatic condition. A similar stratification role for Umod has also been reported in other studies [15, 17, 18].

The results of the multiple regression model showed that FeUmod remained an independent predictor of reduced eGFR even after adjustment for age, sex, and sCrea levels, indicating its autonomous informative value as a risk stratification tool. This aligns with experimental and clinical data highlighting Umod as a marker of nephron integrity and its role in stress adaptation [9, 12, 13].

The clinical significance of biomarker dynamics in response to therapeutic intervention is also noteworthy. In particular, patients receiving GSH or CoQ10 showed favorable changes in uromodulin profile indices, consistent with the antioxidant properties of these agents demonstrated in experimental models [1, 4, 14].

Despite the promising findings, the study has several limitations. First, the relatively small sample size and short observation period do not allow for definitive conclusions regarding the prognostic value of FeUmod. Second, the lack of biopsy-based nephron structure verification limits the interpretation of Umod-related changes. Moreover, the clinical application of such indices requires standardization of laboratory methods and validation in large populations.

In the future, it would be appropriate to develop prognostic models that combine Umod profile indices with classical markers and clinical risk factors to enable personalized risk stratification of patients with CKD [3, 11, 19, 20].

CONCLUSIONS

1. Umod profile indices, particularly FeUmod, demonstrate high diagnostic informativeness in detecting reduced renal filtration capacity in patients with CKD stages 1–3, comparable to classical markers (sCrea and uAlb/uCrea).

2. ROC analysis showed that FeUmod is a sensitive biomarker of functional nephron exhaustion (AUC = 0.985), though it does not significantly outperform sCrea (AUC = 0.977) and uAlb/uCrea (AUC = 0.955) in identifying eGFR < 60 mL/min/1.73 m².

3. Clustering of patients by Umod profile identified three phenotypes that differed in comorbidity burden, anemia, and sex, confirming CKD heterogeneity and the potential of the Umod profile for clinical stratification.

4. FeUmod remained an independent predictor of eGFR after adjustment for age, sex, and sCrea level, supporting its autonomous prognostic significance.

5. The obtained data justify the inclusion of FeUmod in an extended laboratory monitoring panel in nephrology practice for more precise assessment of nephron functional reserve.

REFERENCES

1. Denova, L. D., & Ivanov, D. D. (2024). Assessment of the risk of rapid progression of chronic kidney disease. *Kidneys*, 4(13), 15–21. [in Ukraine]. https://doi.org/10.22141/2307-1257.13.4.2024.480
2. Bai, J., Zhang, L., Di J., et al. (2025). Absence of progression risk of chronic kidney disease in patients with urine protein-creatinine ratio below 500 mg/g: a cohort study with competing risk analysis. *Frontiers in Medicine (Lausanne)*, (12), 1502597. https://doi.org/10.3389/fmed.2025.1502597
3. Kim, J. H., Kim, J., Kim, J., et al. (2025). Risk factors and transitional probability of clinical events in Korean CKD patients using the multistate model. *Scientific Reports*, 15(1), 8582. https://doi.org/10.1038/s41598-025-98168-9
4. Denova, L. D. (2021). Uromodulin as a potential candidate marker for predicting the course of chronic kidney disease. *Kidneys*, 4(10), 71–77. [in Ukraine]. https://doi.org/10.22141/2307-1257.10.4.2021.247898
5. Vonbrunn, E., Ebert, N., Cordasic, N., et al. (2025). Serum Uromodulin as early marker for ischemic acute kidney injury and nephron loss: association with kidney tissue distribution pattern. *Journal of Translational Medicine*, 23(1), 323. https://doi.org/10.1186/s12967-025-06125-x
6. Kaszyńska, A., Kępska-Dzilińska, M., Karakulska-Prystupiuk, E., et al. (2025). Markers of Kidney Injury: Proenkephalin A and Uromodulin, but Not Dickkopf-3, Are Elevated in Patients After Hematopoietic Stem Cell Transplantation. *International Journal of Molecular Sciences*, 26(8), 3581. https://doi.org/10.3390/ijms26083581
7. Carbayo, J., Verdalles, Ú., Díaz-Crespo, F., et al. (2024). Tubular biomarkers in proteinuric kidney disease: histology correlation and kidney prognosis of tubular biomarkers. *Clinical Kidney Journal*, 17(5), sfae146. https://doi.org/10.1093/ckj/sfae146
8. Žeravica, R., Ilinčić, B., Burić, D., et al. (2024). Relationship between serum uromodulin as a marker of kidney damage and metabolic status in patients with chronic kidney disease of non-diabetic etiology. *International Journal of Molecular Sciences*, 25(20), 11159. https://doi.org/10.3390/ijms252011159
9. Nanamatsu, A., Rhodes, G. J., LaFavers, K. A., et al. (2025). Alternative splicing of uromodulin enhances mitochondrial metabolism for adaptation to stress in kidney epithelial cells. *Journal of clinical investigation*, (8), e183343. https://doi.org/10.1172/JCI183343
10. Chang, A. N., Cerutti, G., Ogawa, Y., et al. (2025). Structural basis of human uromodulin filament networks in uropathogen capture. *Structure*, (3), S0969-2126(25)00145-5. https://doi.org/10.1016/j.str.2025.04.011
11. Sun, G., Liu, C., Song, C., et al. (2024). Knowledge mapping of UMOD of English published work from 1985 to 2022: a bibliometric analysis. *International Urology and Nephrology*, 56(1), 249–261. https://doi.org/10.1007/s11255-023-03664-4

12. Si, S., Liu, H., Xu, L., & Zhan, S. (2024). Identification of novel therapeutic targets for chronic kidney disease and kidney function by integrating multi-omics proteome with transcriptome. *Genome Medicine*, 16(1), 84. https://doi.org/10.1186/s13073-024-01356-x
13. Postalcioglu, M., Scherzer, R., Ix, J. H., et al. (2024). Urine epidermal growth factor and kidney function decline in middle-aged adults. *Kidney Medicine*, 6(7), 100846. https://doi.org/10.1016/j.xkme.2024.100846
14. Boima, V., Agyekum, A. B., Ganatra, K., et al. (2025). Advances in kidney disease: pathogenesis and therapeutic targets. *Frontiers in Medicine (Lausanne)*, (12), 1526090. https://doi.org/10.3389/fmed.2025.1526090
15. Strauss-Kruger, M., Olinger, E., Hofmann, P., et al. (2024). *UMOD* Genotype and determinants of urinary uromodulin in African populations. *Kidney International Reports*, 9(12), 3477–3489. https://doi.org/10.1016/j.ekir.2024.09.015
16. Ketteler, M., Evenepoel, P., Holden, R. M., et al. (2025). Conference Participants. Chronic kidney disease-mineral and bone disorder: conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) Controversies Conference. *Kidney Int*, 107(3), 405–423. https://doi.org/10.1016/j.kint.2024.11.013
17. Degenaar, A., Kruger, R., Jacobs, A., & Mels, C. M. C. (2025). Urinary biomarkers of kidney function as predictors of cardiovascular health: a systematic review. *Current Hypertension Reports*, 27(1), 11. https://doi.org/10.1007/s11906-025-01328-5
18. Xiong, L., Wu, C., Chen, S., et al. (2024). Proteomics analysis reveals age-related proteins in the urine of chronic kidney disease patients. *Frontiers in Medicine (Lausanne)*, 2025 (11), 1506134. https://doi.org/10.3389/fmed.2024.1506134
19. Li, R., Tian, S., Liu, J., et al. (2025). Modifiable risk factors and plasma proteomics in relation to complications of type 2 diabetes. *Nature Communications*, 16(1), 2896. https://doi.org/10.1038/s41467-025-57830-6
20. Simeri, A., Pezzi, G., Arena, R., et al. (2025). Artificial intelligence in chronic kidney diseases: methodology and potential applications. *International Urology and Nephrology*, 57(1), 159–168. https://doi.org/10.1007/s11255-024-04165-8

Отримано [Received] 02.06.2025

Оцінка уромодулінового профілю та класичних маркерів у стратифікації ризику прогресування хронічної хвороби нирок 1–3 стадії

Л. Д. Денова, *аспірант,*
ORCID ID: 0000-0002-5678-5885,
marbua18@gmail.com

С. В. Кушніренко, *доктор медичних наук, професор,*
ORCID ID: 0000-0001-5518-7210,
stella-alex@i.ua

Національний університет охорони здоров'я імені П.Л. Шупика, м. Київ, Україна,
Кафедра нефрології і нирковозамісної терапії

Анотація. *У роботі проведено порівняння аналітичної інформативності індексів уромодулінового профілю (фракційної екскреції уромодуліну (FeUmod), фракційної секреції уромодуліну (FsUmod), уромодуліну сироватки(sUmod) та класичних маркерів функції нирок (сироваткового*

креатиніну (sCrea), розрахункової швидкості клубочкової фільтрації (рШКФ) за формулою CKD-EPI, альбумінурії (uAlb) та співвідношення альбумін/креатинін (uAlb/uCrea)) у пацієнтів із хронічною хворобою нирок (ХХН) 1–3 стадії. ***Мета дослідження.*** *Оцінити діагностичну здатність індексів уромодулінового профілю (FeUmod, FsUmod, sUmod) порівняно з класичними маркерами функції нирок (sCrea, рШКФ, uAlb, uAlb/uCrea) у пацієнтів із ХХН 1–3 стадії.* ***Методи і матеріали дослідження.*** *У дослідження включено 91 пацієнта з ХХН 1–3 стадії. Було проаналізовано показники на трьох етапах: T0 (до лікування), T1 (через 3 місяці) та T2 (через 6 місяців). Для оцінки асоціацій проведено кореляційний аналіз. Дискримінаційну здатність щодо зниження рШКФ < 60 мл/хв/1.73 м² визначали методом ROC-аналізу. Кластерний аналіз дозволив виділити фенотипи функціонального стану нирок.* ***Результати дослідження.*** *У кінці дослідження (T2), FeUmod продемонстрував найвищу дискримінаційну здатність, випередивши sCrea, FsUmod, sUmod та uAlb/uCrea. У кореляційному аналізі FeUmod також мав сильний зв'язок із рШКФ. Кластерний аналіз дозволив виділити три фенотипи пацієнтів із відмінним профілем тубулярної і гломерулярної дисфункції.* ***Висновки.*** *Індекси уромодулінового профілю, зокрема FeUmod, перевершують класичні маркери функції нирок за здатністю стратифікувати ризик зниження фільтраційної здатності нирок у пацієнтів з ХХН 1–3 стадії. Доцільним є включення FeUmod до лабораторних інструментів моніторингу в практиці нефролога та інших лікарів.*

Ключові слова: *уромодулін; фракційна екскреція уромодуліну; фракційна секреція уромодуліну; хронічна хвороба нирок; розрахункова швидкість клубочкової фільтрації; стратифікація ризику; біомаркери.*

FUNDING SOURCES

The publication is a part of the PhD research project of the postgraduate student from the Department of Nephrology and Renal Replacement Therapy at the Shupyk National Healthcare University of Ukraine. The research is conducted under the departmental scientific study titled "Uromodulin excretion and its clinical-laboratory evaluation: significance in early diagnosis, renoprotection, and optimization of CKD treatment under molecular stress", within the framework of the state-funded project "Organization and provision of nephrological care under resource limitations and martial law conditions" (state registration number 0123U101260, 2023–2024), and the investigator-initiated project "Study of the impact of hypouricemic therapy in CKD patients and rationale for optimal treatment strategies" (state registration number 0119U101718, 2019–2025).

AUTHOR CONTRIBUTIONS

L. D. Denova – *data collection and analysis, manuscript writing, literature review on the topic, preparation of the manuscript for publication;*
S. V. Kushnirenko – *conceptualization, methodology.*

Автори заявляють про відсутність конфлікту інтересів. [The authors declare no conflict of interest.]

Цитування (ДСТУ):
Денова Л. Д., Кушніренко С. В. Оцінка уромодулінового профілю та класичних маркерів у стратифікації ризику прогресування хронічної хвороби нирок 1–3 стадії. *Лікарська справа*. 2025. № 3. С. 5–14. https://doi.org/10.31640/LS-2025-3-01

Citation (APA):
Denova, L. D, & Kushnirenko, S. V. (2025). Evaluation of the uromodulin profile and classical markers in risk stratification for progression of chronic kidney disease stages 1–3. *Likars'ka Sprava*, (3), 5–14. https://doi.org/10.31640/LS-2025-3-01